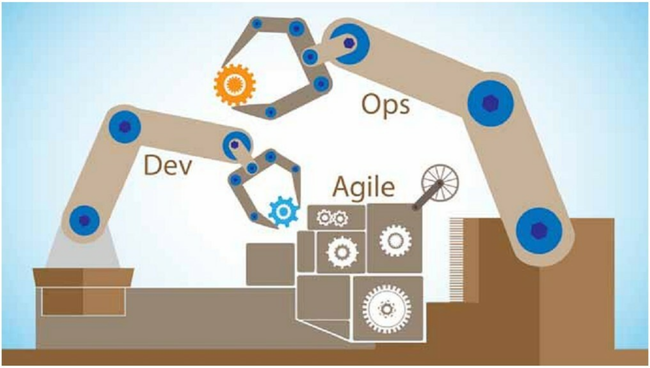
The DevOps Series
Using Docker with Ansible

This article is the eighth in the DevOps series. This month, we shall learn to set up Docker in the host system and use it with Ansible.



Docker provides operating system level virtualisation in the form of containers. These containers allow you to run standalone applications in an isolated environment. The three important features of Docker containers are isolation, portability and repeatability. All along we have used Parabola GNU/Linux-libre as the host system, and executed Ansible scripts on target virtual machines (VM) such as CentOS and Ubuntu.

Docker containers are extremely lightweight and fast to launch. You can also specify the amount of resources that you need such as the CPU, memory and network. The Docker technology was launched in 2013, and released under the Apache 2.0 licence. It is implemented using the Go programming language. A number of frameworks have been built on top of Docker for managing these clusters of servers. The Apache Mesos project, Google's Kubernetes, and the Docker Swarm project are popular examples. These are ideal for running stateless applications and help you to easily scale horizontally.

Setting it up

The Ansible version used on the host system (Parabola GNU/Linux-libre x86_64) is 2.3.0.0. Internet access should be available on the host system. The *ansible/* folder contains the following file:

```
ansible/playbooks/configuration/docker.yml
```

Installation

The following playbook is used to install Docker on the host system:

```
---
- name: Setup Docker
  hosts: localhost
  gather_facts: true
  become: true
  tags: [setup]

  tasks:
    - name: Update the software package repository
      pacman:
        update_cache: yes

    - name: Install dependencies
      package:
        name: "{{ item }}"
        state: latest
```